

Medical History and Risk Assessment

OMFS 512: Integrated Biomedical and Clinical Sciences

Rashidah Wiley, DDS, FAAOMP, D-ABOMP



Mission: To advance the science and art of health through
education, service, scholarship and social accountability

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Learning Outcomes & Competency Statements

Learning Outcomes

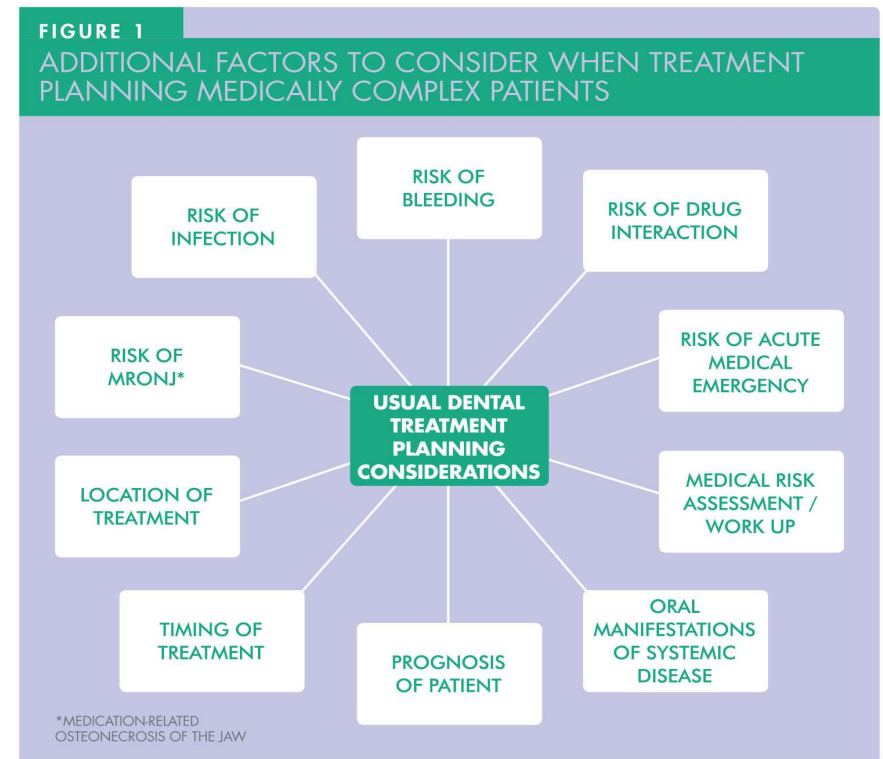
1. Review the different aspects of the medical history and how to apply it to risk assessment.
2. Examine the risk factors associated with dental treatment and systemic diseases.
3. Briefly discuss the modifications associated with treating medically compromised patients.

Competency statements met

1.3, 1.5, and 2.1

Patient Evaluation and Risk Assessment

1. Obtain an accurate medical and social history
2. Review of patient medications
3. Obtain vital signs
4. Physical Examination
5. Order and review diagnostic tests
6. Medical consultations (as needed)
7. Complete risk assessment



Bryant C. Oral surgery: Considerations for the medically complex patient. Prim Dent J. 2022 Sep;11(3):71-79.

Techniques for obtaining Medical History

- A. Medical Model – Interview of patient
- B. Questionnaire

ADA.
American Dental Association
www.ada.org

Health History Form

E-mail: _____ Today's Date: _____

As required by law, our members adhere to written policies and procedures to protect the privacy of information about you that we create, receive or maintain. Your answers to our questions only will and will be kept confidential subject to applicable laws. Please note that you will be asked some questions about your responses to this questionnaire and there may be additional questions concerning your health. This information is vital to allow us to provide appropriate care for you. The office does not use the information to discriminate.

Name	Last	First	Middle	Home Phone	Mobile or cell	Business/Cell Phone	Mobile or cell
	_____	_____	_____	_____ () _____	_____ () _____	_____ () _____	_____ () _____
	City			State		Zip	
Emergency address				Weight	Usual weight	Date of birth	Sex M F
Occupation							
SSN or Patient ID	Emergency Contact			Relationship	Home Phone	Cell Phone	
					_____	_____	

If you are completing this form for another person, what is your relationship to that person? _____

How have you been feeling lately? _____

Do you have any of the following diseases or problems? (Check **OK** if you don't know the answer to the question) Yes No OK

Active tuberculosis	☐	☐	☐
Persistent cough greater than 3 week duration	☐	☐	☐
Cough that produces blood	☐	☐	☐
Have exposure to anyone with tuberculosis	☐	☐	☐

If you answer yes to any of the 4 items above, please stop and return this form to the receptionist.

Dental Information

Please mark (X) your responses to the following questions: Yes No OK

Do your gums bleed when you brush or floss?	☐	☐	☐
Are your teeth sensitive to cold, hot, sweets or pressure?	☐	☐	☐
Does food or floss catch between your teeth?	☐	☐	☐
Is your mouth dry?	☐	☐	☐
Have you ever had any periodontal (gum) treatments?	☐	☐	☐
Have you ever had orthodontics (bracket treatment)?	☐	☐	☐
Have you had any problems associated with previous dental treatment?	☐	☐	☐
Is your home/office supply fluoridated?	☐	☐	☐
Do you drink bottled or filtered water?	☐	☐	☐
If yes, how often? Circle one: DAILY / WEEKLY / OCCASIONALLY	☐	☐	☐
Are you currently experiencing dental pain or discomfort?	☐	☐	☐

What is the reason for your dental visit today? _____

How do you feel about your smile? _____

Medical Information Please mark (X) your response to indicate if you have or have not had any of the following diseases or problems: Yes No OK

Are you now under the care of a physician?	☐	☐	☐
Physician Name: _____	Phone: _____	Mobile or cell: _____	
Address/City/State/Zip: _____			
Are you in good health?	☐	☐	☐
Has there been any change in your general health within the past year?	☐	☐	☐
If yes, what condition is being treated?	☐	☐	☐
Date of last physical exam: _____			

Have you had a serious illness, operation or trauma hospitalized in the past 5 years?	☐	☐	☐
If yes, what was the illness or problem?	☐	☐	☐
Are you taking or have you recently taken any prescription or over the counter medicines?	☐	☐	☐
If yes, please list each including vitamins, natural or herbal preparations and/or diet supplements.	☐	☐	☐

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Form 7-2010

[illegible]

Oral Health/Dental History

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Child Health/Dental History Form

Patient's Name LAST FIRST MIDDLE		DOB	Relationship to Patient
Address STREET CITY STATE ZIP CODE		Phone AREA CODE NUMBER EXTENSION	
Sex ☐ M ☐ F		Date of Birth	

Have you (the parent/guardian) or the patient had any of the following diseases or problems?
 1. Active Tuberculosis, 2. Persistent cough greater than a three-week duration, 3. Cough that produces blood? Q Yes Q No
 If you answer yes to any of the three items above, please stop and return this form to the receptionist.

Has the child had any history of, or conditions related to, any of the following:

☐ Anemia	☐ Cancer	☐ Epilepsy	☐ HIV/AIDS	☐ Mononucleosis	☐ Thyroid
☐ Atrial Fibrillation	☐ Cerebral Palsy	☐ Flushing	☐ Immunizations	☐ Mumps	☐ Tobacco/Drug Use
☐ Asthma	☐ Chicken Pox	☐ Growth Problems	☐ Kidney	☐ Pregnancy (born)	☐ Tuberculosis
☐ Bleeding disorders	☐ Chronic Sinusitis	☐ Hearing	☐ Latex allergy	☐ Rheumatic fever	☐ Venereal Disease
☐ Bone/Joint	☐ Diabetes	☐ Heart	☐ Liver	☐ Seizures	☐ Other
	☐ Ear Aches	☐ Hepatitis	☐ Measles	☐ Sickle cell	

Please list the name and phone number of the child's physician:
 Name of Physician _____ Phone _____

Child's History

1. Is the child taking any prescription and/or over the counter medications or vitamin supplements at this time? Yes No	1. ☐ Yes ☐ No
If yes, please list: _____	
2. Is the child allergic to any medications, i.e. penicillin, antibiotics, or other drugs? If yes, please explain: _____	2. ☐ Yes ☐ No
3. Is the child allergic to anything else, such as certain foods? If yes, please explain: _____	3. ☐ Yes ☐ No
4. How would you describe the child's eating habits? _____	
5. Has the child ever had a serious illness? If yes, when: _____ Please describe: _____	5. ☐ Yes ☐ No
6. Has the child ever been hospitalized? _____	6. ☐ Yes ☐ No
7. Does the child have a history of any other diseases? If yes, please list: _____	7. ☐ Yes ☐ No
8. Has the child ever received a general anesthetic? _____	8. ☐ Yes ☐ No
9. Does the child have any inherited problems? _____	9. ☐ Yes ☐ No
10. Does the child have any speech difficulties? _____	10. ☐ Yes ☐ No
11. Has the child ever had a blood transfusion? _____	11. ☐ Yes ☐ No
12. Is the child physically, mentally, or emotionally handicapped? _____	12. ☐ Yes ☐ No
13. Does the child experience excessive bleeding when cut? _____	13. ☐ Yes ☐ No
14. Is the child currently being treated for any illness? _____	14. ☐ Yes ☐ No
15. Is this the child's first visit to a dentist? If not the first visit, what was the date of the last dentist visit? Date: _____	15. ☐ Yes ☐ No
16. Has the child had any problem with dental treatment in the past? _____	16. ☐ Yes ☐ No
17. Has the child ever had dental radiographs (x-rays) exposed? _____	17. ☐ Yes ☐ No
18. Has the child ever suffered any injuries to the mouth, head or teeth? _____	18. ☐ Yes ☐ No
19. Has the child had any problems with the eruption or shedding of teeth? _____	19. ☐ Yes ☐ No
20. Has the child had any orthodontic treatment? _____	20. ☐ Yes ☐ No
21. What type of water does your child drink? ☐ City water ☐ Well water ☐ Bottled water ☐ Filtered water	
22. Does the child take fluoride supplements? _____	22. ☐ Yes ☐ No
23. Is fluoridated toothpaste used? _____	23. ☐ Yes ☐ No
24. How many times are the child's teeth brushed per day? _____ When are the teeth brushed? _____	24. ☐ Yes ☐ No
25. Does the child suck his/her thumb, fingers or pacifier? _____	25. ☐ Yes ☐ No
26. At what age did the child stop bottle feeding? Age: _____ Breast feeding? Age: _____	
27. Does child participate in or has recreational activities? _____	27. ☐ Yes ☐ No

NOTE: Both doctor and patient are encouraged to discuss any and all relevant patient health issues prior to treatment.
 I certify that I have read and understand the above. I acknowledge that my questions, if any, about inquiries set forth above have been answered to my satisfaction. I will not hold my dentist, or any other member of his/her staff, responsible for any action they take or do not take because of errors or omissions that I may have made in the completion of this form.

Parent/Guardian's Signature _____ Date _____

For completion by dentist

Comments _____

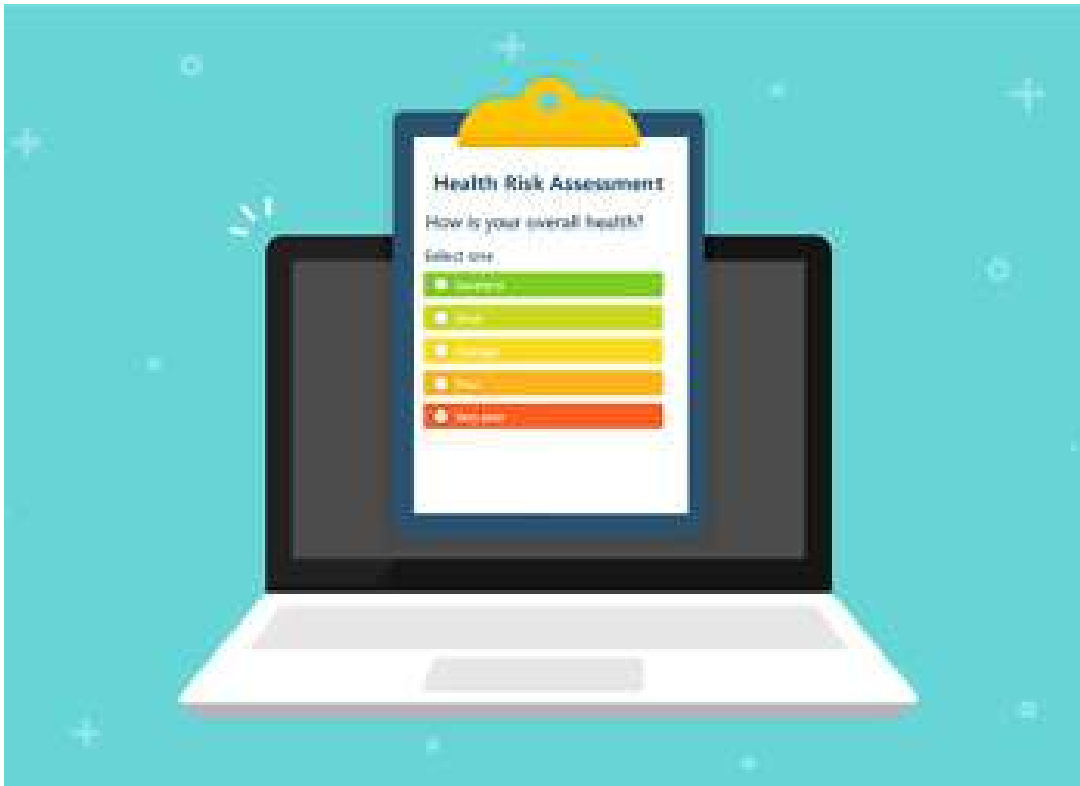
For Office Use Only: ☐ Medical Asst. ☐ Preanesthesia ☐ Radiologist ☐ Anesthesia ☐ Radiology

Date: _____

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Form 8707

To transfer old 1-800-673-6241
or go online at www.ada.org

Patient Evaluation and Risk Assessment



Review Box 1.2 in the Management of the Medically Compromised Patient Textbook

Risk Factor Questions to Consider

1. Type and severity of the disorder
 2. Date of diagnosis
 3. Signs and symptoms associated with the disorder
 4. Is the patient seeing their physicians/specialist on a regular basis?
 5. Treatment
 - a. Current or past treatment for the disorder
 - b. Patient compliance
 - c. Is the disease well or poorly controlled?
 6. Is the patient safe to treat in an outpatient setting?
 7. Routine *versus* emergency dental treatment-**why is this distinction important?**
 8. What complications can occur during or after dental treatment?
 9. Is there an increased risk of bleeding or infection associated with dental treatment?
 10. Are there any risks associated with anesthesia (local and/or general), sedation or nitrous oxide usage?
 11. Is the patient able to tolerate over the counter or prescription medications?
-

Physical Evaluation

- After reviewing the medical history, the dentist should ask supplemental questions to obtain more detailed information about the patient's health status.
- The dentist should also assess non-verbal indicators of the patient's health.
 - **Physical signs of disease**
- Vital signs should be taken at every appointment.
- Observe the patient's behavior and mental cognition.

Communication with Physicians

- Request medical consult NOT medical clearance
- Provide background information
 - What treatment the patient will receive
 - Type of local anesthetic
 - Type of medications the patient will be prescribed after dental treatment
- Be specific with your request
 - Diagnostic tests
 - Ask specific questions regarding health concerns



Medical Risk Assessment

- American Society of Anesthesiology (ASA) Physical Status Score
 - Medical risk assessment which was originally used for anesthesiology
 - Adopted by most healthcare professions
 - Systemic conditions such as hypertension, pulmonary disease, neurologic conditions, hematologic disease and other systemic conditions are taken into consideration.
 - Aids dental clinicians in determining:
 - What type of treatment the patient should or should not receive:
 - Routine vs. Elective Vs Emergency dental therapy
 - Where should the patient receive treatment: outpatient vs hospital facility

ASA Classification

Box 1.4 American Society of Anesthesiologists (ASA) Physical Classifications

ASA I	Normal healthy patient. (E.g., healthy, nonsmoker, no or minimal alcohol use)
ASA II	Patient with mild systemic disease. No significant impact on daily activity; unlikely to have an impact on anesthesia and surgery. (E.g., current smoker, social alcohol drinker, pregnancy, obesity ($30 < \text{BMI} < 40$), well-controlled diabetes mellitus or hypertension, mild lung disease)
ASA III	Patient with moderate to severe systemic disease. Substantive functional limitations to daily activity; probable impact on anesthesia and surgery. (E.g., poorly controlled diabetes mellitus or hypertension, chronic obstructive pulmonary disease, morbid obesity ($\text{BMI} \geq 40$), active hepatitis, alcohol dependence disorder, implanted pacemaker, moderate reduction of ejection fraction, end-stage renal disease (ESRD) undergoing regularly scheduled dialysis, history (>3 months) of MI, cerebrovascular accident (CVA), transient ischemic attack (TIA), or coronary artery disease (CAD)/stents)
ASA IV	Patient with severe systemic disease that is a constant threat to life. Serious limitation of daily activity; likely major impact on anesthesia and surgery. (E.g., recent (<3 months) MI, CVA, TIA, or CAD/stents, ongoing cardiac ischemia or severe valve dysfunction, severe reduction of ejection fraction, respiratory failure requiring mechanical ventilation or mechanical circulation, sepsis, disseminated intravascular coagulation, or ESRD not undergoing regularly scheduled dialysis)
ASA V: a moribund patient not expected to survive without the operation.	
ASA VI: declared brain dead whose organs are being removed for donor purposes.	

Dental Management of the Medically Compromised Dental Patient 10th Edition

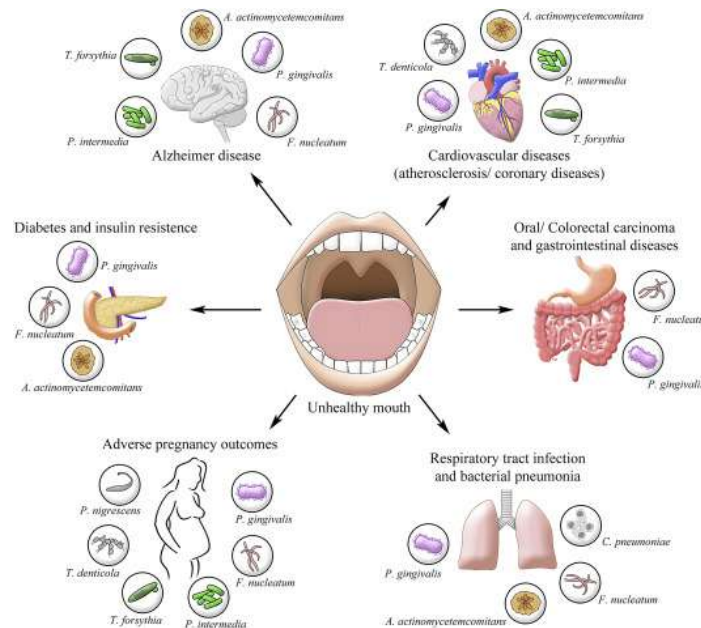
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Impact of Systemic Disease on Oral Health

- There is a significant connection between systemic and oral health
- Signs of systemic disease can manifest in the oral cavity
- Medications taken to control systemic disease can also affect the teeth and oral soft tissues

Risk Factors in Dentistry

- Hypertension/Heart disease
- Impaired Hemostasis
- Susceptibility to infection
- Poor wound healing
- Drug interactions
- Patient's ability to tolerate dental care
 - Ability to physically withstand dental treatment
 - Understanding of treatment and informed consent.



Impaired Hemostasis

- Impaired hemostasis can be due to both congenital and acquired disorders.
 - Medical therapy
 - Herbal supplements
 - Impaired function or destruction of platelets and coagulation factors.



Susceptibility to Infection

- Condition requiring antibiotic prophylaxis
 - Prosthetic (mechanical) heart valves
 - Unrepaired congenital heart defect
 - Previous history of infective endocarditis
 - Organ transplantation
- Impaired immune system
 - Long-term corticosteroid usage
 - Autoimmune conditions
- Diabetes and other endocrine condition

Antibiotic Prophylaxis

Antibiotic Prophylactic Regimens for Dental Procedures

Regimen – Single dose 30 to 60 minutes before procedure

Situation	Agent	Adults	Children
Oral	Amoxicillin	2 g	50 mg/kg
Unable to take oral medication	Ampicillin OR	2 g IM or IV	50 mg/kg IM or IV
	Cefazolin or ceftriaxone	1 g IM or IV	50 mg/kg IM or IV
Allergic to penicillins or ampicillin—oral regimen	Cephalexin*	2 g	50 mg/kg
	OR		
	Azithromycin or clarithromycin	500 mg	15 mg/kg
	OR		
	Doxycycline	100 mg	<45 kg, 2.2 mg/kg >45 kg, 100 mg
Allergic to penicillin or ampicillin and unable to take oral medication	Cefazolin or ceftriaxone†	1 g IM or IV	50 mg/kg IM or IV

Clindamycin is no longer recommended for antibiotic prophylaxis for a dental procedure.
IM indicates intramuscular; and IV, intravenous.

* Or other first- or second-generation oral cephalosporin in equivalent adult or pediatric dosing.

† Cephalosporins should not be used in an individual with a history of anaphylaxis, angioedema, or urticaria with penicillin or ampicillin.

<https://www.contagionlive.com/view/new-recommendations-for-antibiotic-prophylaxis-prior-to-dental-procedures>

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Drug Actions/Interactions

- Most adults in the U.S. are taking at least one type of medication.
- The risk of drug interaction is increased in patients taking multiple medications.
- Patients should be informed about the actions, interactions, and side effects associated with their prescriptions.

Lexicomp

Search Facts and Comparisons eAnswers



New FDA Drug Approvals

[Tebentafusp](#)

[Abrocitinib](#)

[Olopatadine and Mometasone](#)

[Daridorexant](#)

[Levoketoconazole](#)

[More new FDA drug approvals:](#)

[Lexi-Drugs](#)

Special Alerts

> [Buprenorphine: Buprenorphine Safety Alert](#)

> [Buprenorphine and Naloxone: Buprenorphine Safety Alert](#)

> [Casirivimab and Imdevimab: Casirivimab and Imdevimab Emergency Use Authorization and Distribution Update](#)

> [Bamlanivimab and Etesevimab: Bamlanivimab and Etesevimab](#)

References

Clinical Links

[ASHP Drug Shortages](#)

[ISMP \(Institute for Safe Medication Practices\)](#)

[National Library of Medicine](#)

[Orange Book](#)

[Purple Book](#)

[FDA Recalls](#)

Ability to Tolerate Dental Care

- Physiological and psychological considerations
 - Anxiety
 - Pain control
 - Length of the appointment
 - Dental chair position
 - Use of local anesthesia
 - Ability to sign informed consent



Stress Reduction Protocol

Stress reduction considerations

- **Anxiolytic premedication:** benzodiazepine at bedtime night before appointment and 1 hour prior to appointment
- **Appointment scheduling:** early in the day
- **Minimize waiting time:** in waiting room and dental chair
- **Preoperative and postoperative vital signs:** blood pressure, heart rate and rhythm, respiratory rate, pain score
- **Sedation during treatment:** iatrosedation (music and video distraction, hypnosis), nitrous oxide/oxygen analgesia or pharmacosedative procedures including oral, inhalational, intramuscular, intranasal or intravenous (minimal or moderate) sedation, or general anesthesia
- **Treatment duration:** short appointments

The ADA Practical Guide to Patients with Medical Conditions

Dental Modifications

- Dental modifications are made based on the patient's medical condition(s).
 - Review the health history prior to the appointment
 - Obtain baseline vitals prior to treatment
 - Obtain INR values in patients on Coumadin prior to surgical procedures
 - Stress reduction protocol
 - Nitrous oxide and other forms of sedation to help with anxiety
 - Limited local anesthesia use
 - Use hemostatic agents to control bleeding
 - Avoid medications that will increase bleeding
 - Antibiotic prophylaxis for susceptible patients
 - Emergency kit containing medications to help with medical complications.

Resources

